OPUS TECHNOLOGIES

Interview Preparation & Guidance Guide

Based on My Experience

4 Rounds • All Rounds Elimination-Based

Important: This guide is based on my experience with the Opus Technologies selection process. The exact questions, difficulty, and process may change from year to year. Use this as a preparation guide, not as a guaranteed question list.

1. Selection Process

Round	Mode	Round Type	Elimination
Round 1	Online	Aptitude & Reasoning	Yes
Round 2	Online	Technical Assessment	Yes
Round 3	In-person	Technical Interview	Yes
Round 4	In-person	HR Interview	Yes

2. Round 1 — Aptitude & Reasoning

The first round primarily tested logical reasoning, quantitative aptitude, verbal ability, and visual/spatial reasoning.

Topics Experienced

Logical & Visual Reasoning

- Mirror Images — identify the correct mirror image of a figure.
- Rotation / Spatial Orientation — rotate an image or figure, including 180° rotation, and identify the correct option.
- Paper Folding — predict the final pattern after folding and unfolding paper.
- Cube / Figure Completion — use multiple views of a cube or figures to identify the missing/matching figure.
- Pattern & Figure-Based Puzzles — identify the next or matching figure based on a visual pattern.
- Coding & Decoding — identify relationships between letters, numbers, or symbols.
- Blood Relations & Family Tree — determine relationships from given information.

Quantitative Aptitude

- Percentages
- Profit & Loss
- Data Interpretation

Verbal Ability

- Reading comprehension / answering questions based on a paragraph
- Sentence rearrangement / Para Jumbles

Preparation Advice

- Practice with a timer to improve speed and accuracy.
- Learn common patterns and shortcuts where appropriate.
- Avoid spending too much time on one difficult question.
- Practice visual reasoning separately because figure-based questions can be time-consuming.

3. Round 2 — Technical Assessment

The second round consisted of a technical assessment with coding problems, MCQs, output-based questions, debugging/error identification, and Computer Science fundamentals.

Coding Problems

- Count vowels in a string
- Check whether a string is a palindrome

- Print the Fibonacci series
- Check whether a number is prime
- Find factorial of a number
- Reverse a string
- Reverse a number
- Find the largest/smallest element in an array
- Find the second-largest element in an array
- Count character frequency in a string
- Check whether two strings are anagrams
- Find the sum of digits of a number
- Check whether a number is an Armstrong number
- Remove duplicate elements from an array
- Find duplicate elements in an array
- Count words in a sentence
- Sort an array
- Search for an element in an array

Debugging & Error Identification

- Syntax errors
- Logical errors
- Runtime errors
- Incorrect conditions
- Loop errors
- Array/index errors
- Variable and function errors
- Incorrect output

Output-Based Programming

- Loops and nested loops
- if/else conditions
- Functions
- Arrays and strings
- Operators
- Increment/decrement
- Recursion basics
- Scope
- Object-oriented concepts

Technical MCQs

Operating Systems

- Process vs Thread
- CPU Scheduling
- Deadlock
- Memory Management
- Virtual Memory
- Paging
- Synchronization

- Context Switching
- Basic Linux concepts

DBMS

- Primary Key
- Foreign Key
- Candidate Key
- Normalization
- SQL basics
- Joins
- GROUP BY
- ORDER BY
- Aggregate functions
- WHERE vs HAVING
- Transactions
- ACID properties
- Indexing

Computer Networks

- OSI Model
- TCP/IP Model
- TCP vs UDP
- HTTP vs HTTPS
- IP Address
- DNS
- MAC Address
- Ports
- Client-server architecture

OOP

- Class & Object
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Method Overloading
- Method Overriding
- Constructors

Programming Fundamentals

- Variables & data types
- Operators
- Conditional statements
- Loops
- Functions
- Arrays
- Strings
- Recursion basics
- Exception handling

AI/ML Basics

- AI vs ML
- Supervised vs Unsupervised Learning
- Classification vs Regression
- Training vs Testing data
- Overfitting vs Underfitting
- Basic ML algorithms
- Accuracy, Precision, Recall
- Basic neural-network concepts

Tip: For coding questions, do not only memorize solutions. Be able to explain the input, logic, output, edge cases, and time/space complexity.

4. Round 3 — In-Person Technical Interview

This was a deep technical and resume-based discussion. The interviewer could build follow-up questions directly from your answers, skills, projects, internship, and resume.

1) Coding Question

I was first asked to write a coding problem of my choice and explain it. Follow-up questions included:

- Why did you choose this problem?
- Why did you use this approach?
- How does your solution work?
- Can you optimize it?
- What is the time complexity?
- What is the space complexity?
- What happens for edge cases?
- Can you solve it using another approach?

Best practice: For every coding problem you practice, be ready to explain Problem → Approach → Code → Example → Complexity → Optimization.

2) SQL Query

Prepare especially:

- SELECT, WHERE, GROUP BY, HAVING, ORDER BY
- INNER JOIN, LEFT JOIN, RIGHT JOIN
- COUNT(), SUM(), AVG(), MAX(), MIN(), DISTINCT
- Subqueries
- Queries involving multiple tables

3) Resume-Based Questions

Anything written on your resume can become an interview question. If you mention a skill, technology, tool, internship, or project, be prepared to explain it.

Example: I had mentioned Linux, Git, and GitHub, so I was asked about Linux and Git commands.

4) Projects — Deep Discussion

- What is your project?
- Why did you build it?

- What problem does it solve?
- What was your role?
- What technologies did you use?
- Why did you choose this technology?
- Why did you use this API?
- How does the API work?
- How does the frontend communicate with the backend?
- How is the database structured?
- What challenges did you face?
- How did you solve them?
- What would you improve?

Golden rule: Don't memorize your project description. Understand your project deeply enough to answer repeated 'why?' and 'how?' questions.

5) Internship Experience

- What did you work on?
- What was your responsibility?
- What technologies did you use?
- What did you learn?
- What problems did you face?
- How did you solve them?
- What was your contribution?

6) Linux Basics

If Linux is mentioned on your resume, know the purpose and basic usage of:

pwd ls cd mkdir rm cp mv touch cat grep find chmod ps top kill

7) Git & GitHub Basics

Know the purpose and basic usage of:

git init git clone git status git add git commit git push git pull git fetch git branch git checkout git merge git log

- Repository
- Branch
- Commit
- Merge
- Pull vs Fetch
- Local vs Remote repository
- Merge conflicts

5. Round 4 — HR Interview

The final round focused mainly on personality, communication, motivation, behavioral situations, and fit.

Common Questions

About Yourself

- Tell me about yourself.
- Tell me something that is not mentioned on your resume.
- What are your strengths?
- What are your weaknesses?
- Where do you see yourself in 5 years?

Projects & Internship

- Why did you choose this project?
- What did you learn from your internship?
- What was your biggest challenge?
- What would you do differently?

Company & Career

- Why Opus Technologies?
- Why should we hire you?
- Why do you want this role?
- What are your career goals?

Relocation

- Are you comfortable relocating?
- Are you comfortable working from another location?
- Are you comfortable with the company's work requirements?

Situational / Behavioral

- How would you handle a conflict with a teammate?
- What would you do if your teammate wasn't contributing?
- What would you do if you disagreed with your manager?
- How would you handle a tight deadline?
- What would you do if you made a mistake?
- How do you handle failure?
- Tell me about a difficult situation you faced and how you handled it.

How to Answer Scenario-Based Questions

Use a simple structure: Situation → Action → Reasoning → Result. Avoid vague answers such as 'I will communicate and solve the problem.' Give a clear, logical approach and, when possible, a real example.

6. Final Preparation Checklist

Aptitude

- ☐ Percentages
- ☐ Profit & Loss
- ☐ Data Interpretation
- ☐ Logical Reasoning
- ☐ Coding & Decoding
- ☐ Blood Relations

- ☐ Mirror Images
- ☐ Paper Folding
- ☐ Cube/Spatial Reasoning
- ☐ Sentence Rearrangement
- ☐ Reading Comprehension

Coding

- ☐ Strings
- ☐ Arrays
- ☐ Numbers
- ☐ Loops
- ☐ Functions
- ☐ Basic Recursion
- ☐ Palindrome
- ☐ Fibonacci
- ☐ Prime Numbers
- ☐ Sorting/Search
- ☐ Frequency problems
- ☐ Basic debugging
- ☐ Output prediction

CS Fundamentals

- ☐ OOP
- ☐ DBMS
- ☐ SQL
- ☐ OS
- ☐ Computer Networks
- ☐ Programming Fundamentals
- ☐ Basic AI/ML

Interview

- ☐ Explain one coding problem
- ☐ Explain time & space complexity
- ☐ Write SQL queries
- ☐ Explain every resume skill
- ☐ Explain every project
- ☐ Explain internship
- ☐ Explain why particular technologies/APIs were used
- ☐ Linux basics
- ☐ Git/GitHub basics

HR

- ☐ Tell me about yourself
- ☐ Why Opus?
- ☐ Strengths/weaknesses
- ☐ Relocation
- ☐ Conflict scenarios
- ☐ Teamwork

- ☐ Failure/challenges
- ☐ Career goals
- ☐ Something not on your resume

7. Final Advice to Juniors

Don't prepare only for the questions. Prepare to explain your thinking.

Opus can ask you something from your coding solution, resume, project, internship, or skills and continue asking 'why?' and 'how?'. Your goal should not be to memorize answers. Your goal should be to understand what you have written and what you have built.

Good luck — prepare smart, understand your fundamentals, and stay calm.

Best of luck, everyone!

— Ayush Chainani
2026 Batch | Alumni

Prepare. Practice. Believe. Deliver.